Program Paper

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1 Goal (Current Thought)

1. Show that prolongated associativity operad $\mathbb{P}\mathscr{M}$ is well-defined as a G-operad.

2. Promoe $\mathbb{P}\mathscr{M}$ promotes to a well-defined permutativity G-operad in Cat_G (say $\mathscr{E}\mathbb{P}\mathscr{M}$), and use it to define the new pseudoalgebras.

3. Show that $\mathscr{F}_{G,\mathbb{R}}$ pseudofunctors (if possible, G-functors, meaning preserving morphism sets' G equivariance) is well-defined, together with isomorphism to new pseudoalgebras as above.

4. (Not sure if it's true), but if we can show there is an equivalence of G-categories between the original $\mathscr{P}_G$ and $\mathscr{E}\mathbb{P}\mathscr{M}$, then maybe we cn also show the equivalence from the original categories.

The first three goals based on intuition and inspection seems to be true.

2 Introduction (ND)

3 Categorical Notations (Keep adding, like statements on morphisms)

(Main references are Peter's papers).

Recall that $\mathscr{F}$ stands for the category of based finite sets, with objects being $\mathbf{n} := \{0, 1, ..., n\}$ (with 0 as basepoint), and homset $\mathscr{F}(\mathbf{n}, \mathbf{m})$ as all based maps between them. For generalization to equivariance setting for a fixed finite group G, one needs an extra G-equivariant structure.

Definition 3.1. Let $\mathscr{F}_G$ denotes the G-category of based finite G-sets, with objects being $\mathbf{n}^\alpha := \{0, 1, ..., n\}$ as set, together with a group homomorphism $\alpha : G \to \Sigma_n$; the homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta) = \mathscr{F}(\mathbf{n}, \mathbf{m})$ as based set maps.

Here, we view $\Sigma_n \subseteq \mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{n}^{\alpha'})$ as permutation of the nonzero elements in $\mathbf{n}$. Which, the action on the homset $\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)$ is given by conjugation

$$g \cdot f := \beta(g) \circ f \circ \alpha(g)^{-1}$$

We'll use $G\mathscr{F} := [\mathscr{F}_G]^G$ as the G-fixed subcategory, which have the same objects as $\mathscr{F}_G$, while the homset $G\mathscr{F}(\mathbf{n}^\alpha, \mathbf{m}^\beta) = [\mathscr{F}_G(\mathbf{n}^\alpha, \mathbf{m}^\beta)]^G$ are all the based G-maps between the two.

For the case $\mathscr{F}$, there are several subcategories, including Π (with all compositions of injections and projections), Λ (with all injections), $\Lambda^\perp$ (with all projections), Σ (with only bijections), and $\mathscr{E}$ (the effective maps). We make similar definition for $\mathscr{F}_G$:

Definition 3.2. Let $\Pi_G \subset \mathscr{F}_G$ be the subcategory where each morphism $\phi : \mathbf{n}^\alpha \to \mathbf{m}^\beta$ has $|\phi^{-1}(j)| = 0$ or 1 for all $0 \neq j \in \mathbf{m}$.

Let $\Lambda_G \subset \Pi_G$ be the subcategory where each ϕ is injective (called *injections*); conversely, let $\Lambda_G^\perp \subset \Pi_G$ stands for the subcategory where each ϕ is surjective (called *projections*).

Let $\Sigma_G = \Lambda_G \cap \Lambda_G^\perp$, which is a subcategory with all ϕ being bijections.

Let $\mathscr{E}_G \subset \mathscr{F}_G$ be the subcategory where all morphism satisfies $\phi^{-1}(0) = \{0\}$, which these morphisms are called *effective*.

Similarly, we use $G\Pi, G\Lambda, G\Lambda^\perp$, and $G\mathscr{E}$ to denote the G-fixed subcategory for each (which homsets are limited to G-maps).

Insert some theorems (probably do some brief verification on them), keep increase them if needed.

However, there is an important notation we need:

Notation 3.3. Given a based finite G-set $\mathbf{n}^\alpha$, denote the group $\Sigma_{\mathbf{n}^\alpha} := G\Sigma(\mathbf{n}^\alpha, \mathbf{n}^\alpha)$, all the G-automorphisms of $\mathbf{n}^\alpha$.

4 Equivariance Preliminaries (Keep adding)

Given a group Π and element $\sigma \in \Pi$, we let $c_\sigma \in \mathrm{Aut}(\Pi)$ denotes the conjugation by σ, $c_\sigma(\psi) := \sigma\psi\sigma^{-1}$.

Definition 4.1. Fix a finite group G. Given Σ_n together with a group homomorphism $\alpha : G \to \Sigma_n$, the *semidirect product* $\Sigma_n \rtimes_{\alpha_c} G$ is the group with underlying set $\Sigma_n \times G$, together with product defined using G's conjugation action on Σ_n using α, as follow:

$$(\sigma, g) \cdot (\sigma', g') := (\sigma\ \alpha(g)\ \sigma'\ \alpha(g)^{-1}, gg')$$

Given a based finite G-set $\mathbf{n}^\alpha$, we ought to extend it to a semidirect product action:

Lemma 4.2. *The left G-action on $\mathbf{n}^\alpha$ extends to a $(\Sigma_n \rtimes_{\alpha_c} G)$-action, given by*

$$(\sigma, g)(i) := \sigma \circ \alpha(g)(i)$$

Proof. Fix $(\sigma, g), (\sigma', g') \in \Sigma_n \rtimes_{\alpha_c} G$, computation shows

$$(\sigma, g)[(\sigma', g')(i)] = (\sigma, g)[\sigma' \circ \alpha(g')(i)] = \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g')(i)$$

$$\begin{aligned}
[(\sigma, g) \cdot (\sigma', g')](i) &= (\sigma\ \alpha(g)\ \sigma'\ \alpha(g)^{-1}, gg')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g)^{-1} \circ \alpha(gg')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g)^{-1} \circ \alpha(g) \circ \alpha(g')(i) \\
&= \sigma \circ \alpha(g) \circ \sigma' \circ \alpha(g')(i) \\
&= (\sigma, g)[(\sigma', g')(i)]
\end{aligned}$$

Also, it's clear identity acts trivially, verifying it's a group action. $\square$

Another important notion is the twisted G-action if the space is given both G and Σ_n-action. For this, consider the following modification:

Definition 4.3. Given a group homomophism $\alpha : G \to \Sigma_n$, its *graph subgroup*

$$\Gamma_\alpha := \{(g, \alpha(g)) \in G \times \Sigma_n\}$$

Using the first projection $\pi_1 : G \times \Sigma_n \twoheadrightarrow G$, it restricts to an isomorphism $\Gamma_\alpha \cong G$.

Given a $(G \times \Sigma_n)$-set X, define the *twisted G-action* on X by

$$g \cdot_\alpha x := (g, \alpha(g)) \cdot x$$

Which, the set with such twisted action is denoted as X^α.

Generalize the definition to arbitrary categories?

For further purpose, we consider the semidirect product action on each finite symmetric group:

Definition 4.4. Given Σ_n and a group homomorphism $\alpha : G \to \Sigma_n$, define a right $(\Sigma_n \rtimes_{\alpha_c} G)$-action on Σ_n

5 G-Operads

5.1 Composability of Finite G-Sets

Here, each $\mathbf{j}^\alpha \in \mathscr{F}_G$ is a based finite set $\mathbf{j} = \{0, 1, ..., j\}$, together with a group homomorphism $\alpha : G \to \Sigma_n$.

Definition 5.1. The tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$ is called *composable*, if given any $g \in G$ and $r, s \in \{1, ..., k\}$, $\beta(g)(r) = s$ implies $\mathbf{j}_r^{\alpha_r} = \mathbf{j}_1^{\alpha_r}$.

In this case, denote $j := \sum_{r=1}^k j_r$, define the function $\gamma := \gamma(\beta; \vee_r \alpha_r) : G \to \Sigma_j$ by

$$\gamma(g)(i) := \alpha_s(g)(i)$$

where $\beta(g)(r) = s$ (permutation of blocks), $i \in \mathbf{j}_r$ (as the rth block in $\mathbf{j}$), and $\alpha_s(g)(i) \in \mathbf{j}_s$ (as the sth block in $\mathbf{j}$).

The composability is for composing the group homomorphisms into a larger group homomorphism:

Proposition 5.2. *If the tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$ is composable, then $\gamma(\beta; \vee_r \alpha_r)$ is a group homomorphism.*

Proof. Fix $r \in \{1, ..., k\}$, $g, h \in G$, let $s := \beta(g)(r)$, and $t = \beta(h)(s) = \beta(hg)(r)$. Now, the computation shows the following for $i \in \mathbf{j}_r^{\alpha_r}$, where r, i are arbitrary:

$$\gamma(g)(i) = \alpha_s(g)(i) \in \mathbf{j}_s$$

$$\gamma(h)(\alpha_s(g)(i)) = \alpha_t(h)(\alpha_s(g)(i)) = \alpha_t(h)(\alpha_t(g)(i)) = \alpha_t(hg)(i) \in \mathbf{j}_t$$

here we use the assumption where $\mathbf{j}_r^{\alpha_r} = \mathbf{j}_s^{\alpha_s} = \mathbf{j}_t^{\alpha_t}$. This indicates

$$\gamma(h) \circ \gamma(g)(i) = \gamma(h)(\alpha_s(g)(i)) = \alpha_t(hg)(i) = \gamma(hg)(i)$$

concluding $\gamma(h) \circ \gamma(g) = \gamma(hg)$. $\square$

Here, we use $\epsilon_j : G \to \Sigma_j$ to denote the trivial group homomorphism. It satisfies unit axioms:

Proposition 5.3. *Given the composable tuples $(\mathbf{j}^\alpha; \mathbf{1}_1^{\epsilon_{1,1}}, ..., \mathbf{1}_j^{\epsilon_{1,j}})$ (with j copies of $\mathbf{1}$) and $(\mathbf{1}^{\epsilon_1}; \mathbf{j}^\alpha)$, then*

$$\gamma(\alpha; \vee_t \epsilon_{1,t}) = \alpha = \gamma(\epsilon_1; \alpha) : G \to \Sigma_j$$

Proof. For the first equality, for any $r \in \{1, ..., j\}$ and $g \in G$, let $s := \alpha(g)(r)$ (which, r, s can be viewed as the only element in the rth and sth block respectively, which we denote $1_r \in \mathbf{1}_r$, $1_s \in \mathbf{1}_r$ respectively). Then, the following equality holds:

$$\gamma(\alpha; \vee_t \epsilon_{1,t})(g)(1_r) = \epsilon_1(g)(1_s) = 1_s$$

Swapping back to normal labels, we get $\gamma(\alpha; \vee_t \epsilon_{1,t})(g)(r) = s$.

For the second equality, since there's only one block involved, G acts trivially on the block, hence $\gamma(\epsilon; \alpha)(g)(r) = \alpha(g)(r) = s$. The two equalities concluded $\gamma(\alpha; \vee_t \epsilon_{1,t})(g) = \alpha(g) = \gamma(\epsilon_1; \alpha)(g)$. $\square$

Finally, it also satisfies associativity:

Proposition 5.4. *Given composable tuple* $(\mathbf{k}^{\delta}; \mathbf{j}_1^{\beta_1}, ..., \mathbf{j}_k^{\beta_k})$*, and each* $r \in \{1, ..., k\}$ *is associated with a composable tuple* $(\mathbf{j}_r^{\beta_r}; \mathbf{i}_{r,1}^{\alpha_{r,1}}, ..., \mathbf{i}_{r,j_r}^{\alpha_{r,j_r}})$ *(with* $\delta(g)(r) = t$ *implies the tuples are identical), then*

$$\gamma\left(\gamma(\delta; \vee_r \beta_r); \bigvee_r (\vee_s \alpha_{r,s})\right) = \gamma\left(\delta; \bigvee_r \gamma(\beta_r; \vee_s \alpha_{r,s})\right) : G \to \Sigma_i$$

Here, $\bigvee_r (\vee_s \alpha_{r,s})$ is wedging the $\mathbf{i}_{r,s}$ blocks in the order

$$\mathbf{i}_{1,1}, \mathbf{i}_{1,2}, ..., \mathbf{i}_{1,j_1}; \mathbf{i}_{2,1}, \mathbf{i}_{2,2}, ..., \mathbf{i}_{2,j_2}; ...; \mathbf{i}_{k,1}, \mathbf{i}_{k,2}, ..., \mathbf{i}_{k,j_k}$$

Also, observe that if $\delta(g)(r) = t$, then $\gamma(\beta_r; \vee_s \alpha_{r,s}) = \gamma(\beta_t; \vee_s \alpha_{t,s})$ (as we assume the tuples for r, t are identical), so the right map makes sense.

Proof. Denote the left map as γ_1, the right map as γ_2.

Given $\ell_{r,s} \in \mathbf{i}_{r,s}$ (r indicates the block for $\mathbf{j}$'s, and s indicates the block for $\mathbf{i}_r$'s), $g \in G$, suppose $\delta(g)(r) = t$ (implying $\mathbf{j}_r^{\beta_r} = \mathbf{j}_t^{\beta_t}$), and $\beta_r(g)(s) = \beta_t(g)(s) = w$, we get

$$\gamma(\delta; \vee_r \beta_r)(g)(r, s) = (t, \beta_t(g)(s)) = (t, w)$$

(Here, r indicates the rth block in $\mathbf{j}$, s indicates the sth element in $\mathbf{j}_r$).

Hence, γ_1 yields the following computation:

$$\gamma_1(g)(\ell_{r,s}) = \alpha_{t,w}(\ell_{t,w})$$

Similarly, if consider $\gamma(\beta_t; \vee_u \alpha_{t,u})(g)(\ell_{t,s}) = \alpha_{t,w}(g)(\ell_{t,w})$, hence

$$\gamma_2(g)(\ell_{r,s}) = \gamma(\beta_t; \vee_u \alpha_{t,u})(g)(\ell_{t,s}) = \alpha_{t,w}(\ell_{t,w})$$

due to the assumption $\delta(g)(r) = t$ for permuting the $\mathbf{j}$'s blocks. This indicates $\gamma_1(g) = \gamma_2(g)$ for all g. $\square$

5.2 G-Operad's Axioms

Instead of indexing over finite sets (objects in $\mathscr{F}$), the concept of G-operad is instead indexed over finige G-sets (objects in $\mathscr{F}_G$). Here, we fix $(\mathscr{V}, \otimes, \kappa)$ as our symmetric monoidal category.

Here the definitions are based on my personal interpretation, need verification!

Definition 5.5. A *G-operad* $\mathscr{C}_G$ in $\mathscr{V}$ consists of objects $\mathscr{C}_G(\mathbf{j}^\alpha)$ with a left G-action and a right Σ_j-action that combines to a right $(\Sigma_j \rtimes_{\alpha_c} G)$-action, unit morphism $\eta : \kappa \to \mathscr{C}_G(\mathbf{1}^{\epsilon_1})$, and structure morphism

$$\gamma_G : \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^{k} \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \to \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)})$$

for each composable tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$. They must satisfy associative, unital, and equivariance in the following sense:

(a) Given composable tuple $(\mathbf{k}^\delta; \mathbf{j}_1^{\beta_1}, ..., \mathbf{j}_k^{\beta_k})$, and each $r \in \{1, ..., k\}$ provides another composable tuple $(\mathbf{j}_r^{\beta_r}; \mathbf{i}_{r,1}^{\alpha_{r,1}}, ..., \mathbf{i}_{r,j_r}^{\alpha_{r,j_r}})$ (with assumption $\delta(g)(r) = t$ implies the composable tuple for $\mathbf{j}_r^{\beta_r}$ and $\mathbf{j}_t^{\beta_t}$ are identical), let $j := \sum_r j_r$, $h_r := \sum_s i_{r,s}$, and $i := \sum_r h_r$, the following associativity diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\beta_r})\right] \otimes \left[\bigotimes_{s=1}^{j_r} \mathscr{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right] \arrow[r, "\gamma_G \otimes \mathrm{id}"] \arrow[dd, "\text{shuffle}"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\delta;\vee_r\beta_r)}) \otimes \left[\bigotimes_{r=1}^k \bigotimes_{s=1}^{j_r} \mathcal{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right] \arrow[d, "\gamma_G"] \\
 & \mathscr{C}_G(\mathbf{i}^{\gamma^2}) \\
\mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \left[\mathscr{C}_G(\mathbf{j}_r^{\beta_r}) \otimes \left(\bigotimes_{r=1}^{j_r} \mathscr{C}_G(\mathbf{i}_{r,s}^{\alpha_{r,s}})\right)\right]\right] \arrow[r, "\mathrm{id}\otimes(\gamma_G)^k"'] & \mathscr{C}_G(\mathbf{k}^\delta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{h}_r^{\gamma(\beta_r;\vee_s\alpha_{r,s})})\right] \arrow[u, "\gamma_G"']
\end{tikzcd}$$

Here, $\gamma^2 := \gamma\left(\gamma(\delta; \vee_r\beta_r); \bigvee_r (\vee_s\alpha_{r,s})\right) = \gamma\left(\delta; \bigvee_r \gamma(\beta_r; \vee_s\alpha_{r,s})\right)$ based on **Proposition 3.4**.

(b) The following unit diagrams commute:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{j}^\alpha) \otimes \kappa^j \arrow[r, "\cong"] \arrow[d, "\mathrm{id}\otimes\eta^j"'] & \mathscr{C}_G(\mathbf{j}^\alpha) \\
\mathscr{C}_G(\mathbf{j}^\alpha) \otimes [\mathscr{C}_G(\mathbf{1}^{\epsilon_1})]^j \arrow[ur, "\gamma_G"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\kappa \otimes \mathscr{C}_G(\mathbf{j}^\alpha) \arrow[r, "\cong"] \arrow[d, "\eta\otimes\mathrm{id}"'] & \mathscr{C}_G(\mathbf{j}^\alpha) \\
\mathscr{C}_G(\mathbf{1}^{\epsilon_1}) \otimes \mathscr{C}_G(\mathbf{j}^\alpha) \arrow[ur, "\gamma_G"'] &
\end{tikzcd}$$

(c) Given composable tuple $(\mathbf{k}^\beta; \mathbf{j}_1^{\alpha_1}, ..., \mathbf{j}_k^{\alpha_k})$, the following $\Sigma_{\mathbf{k}^\beta}$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\mathrm{id}\otimes\sigma"] \arrow[d, "\sigma\otimes\mathrm{id}"'] & \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_{\sigma^{-1}(r)}^{\alpha_{\sigma^{-1}(r)}})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\gamma_G"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)})
\end{tikzcd}$$

(d) Similar notation, the following $(\Sigma_k \rtimes_{\beta_c} G)$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "(\sigma,g)\otimes\mathrm{id}"] \arrow[d, "\gamma_G"'] & \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}) \arrow[r, "(\sigma(j_1,\ldots,j_r),g)"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)})
\end{tikzcd}$$

(e) The following $\big((\Sigma_{j_1} \times \ldots \times \Sigma_{j_k}) \rtimes_{(\alpha_1,\ldots,\alpha_k)_c} G\big)$-equivariance diagram commutes:

$$\begin{tikzcd}
\mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[r, "\mathrm{id}\otimes((\tau_1,\ldots,\tau_k),g)"] \arrow[d, "\gamma_G"'] & \mathscr{C}_G(\mathbf{k}^\beta) \otimes \left[\bigotimes_{r=1}^k \mathscr{C}_G(\mathbf{j}_r^{\alpha_r})\right] \arrow[d, "\gamma_G"] \\
\mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)}) \arrow[r, "(\tau_1\oplus\ldots\oplus\tau_k,g)"'] & \mathscr{C}_G(\mathbf{j}^{\gamma(\beta;\vee_r\alpha_r)})
\end{tikzcd}$$

5.3 Prolongation of Associativity G-Operad, and Permutativity G-Operad

Definition 5.6. Define the *associativity G-operad* $\mathscr{M}_G$ in the category **GSet**, by $\mathscr{M}_G(\mathbf{n}^\alpha) := (\Sigma_n)^\alpha$, endows with the right action defined in **Equivariance Preliminaries**. Which, $\mathscr{M}_G = \mathbb{P}\mathscr{M}$ if consider prolongation.

Definition 5.7. Define the new *permutativity G-operad* $\mathscr{P}'_G$ in the category of small G-categories $\mathbf{Cat}_G$, as the chaotic prolongation